In the first two cases, the critical angle i_c for the material of the prism must be less than 45°. Prism in Figure 6.24(b) inverts the object on the same side. Prism in Figure 6.24(c) inverts the object on the other side.

6.4.7.4 Radius of illumination (Snell's window)



Figure 6.25 Light source inside water tank

When a source of light like electric bulb is kept inside a water tank, the light from the source travels in all direction inside the water. The light that is incident on the water surface at an angle less than the critical angle will undergo refraction and emerge out from the water. The light incident at an angle greater than critical angle will undergo total internal reflection.The light falling particularly at critical angle graces the surface. Thus, the entire surface of water appears illuminated when seen from outside as shown in Figure 6.25.

On the other hand, when the light entering the water from outside is seen from inside the water, the view is restricted to a particular angle equal to the critical angle i_c. The restricted illuminated circular area is called *Snell's window* as shown in Figure 6.26(a).The Figure 6.26(b) shows the angle of view for water animals.

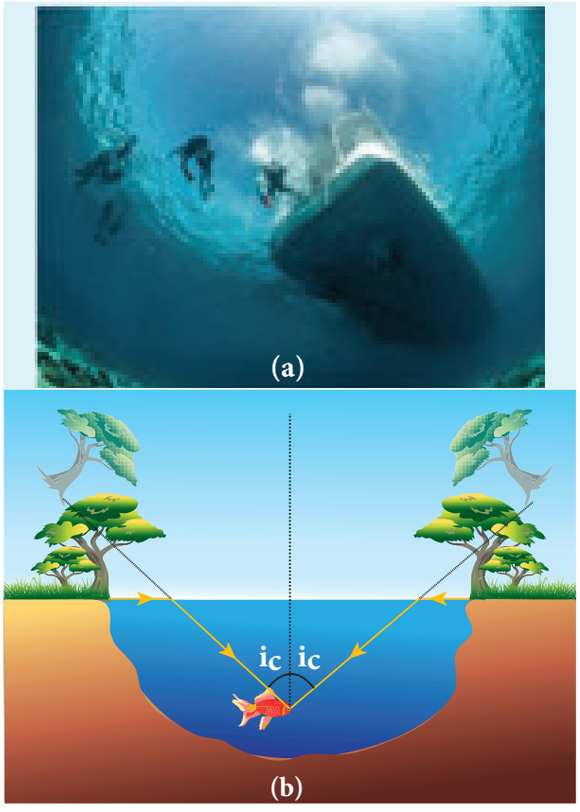


Figure 6.26 (a) Snell's window and (b) angle of view for water animals

The angle of view for water animals is restricted to twice the critical angle $2i_c$. The critical angle for water is 48.6°. Thus the angle of view is 97.2°. The radius R of the circular area depends on the depth d from which it is seen and also the refractive index n of the medium. The radius R of Snell's window can be deduced with the illustration as shown in Figure 6.27.

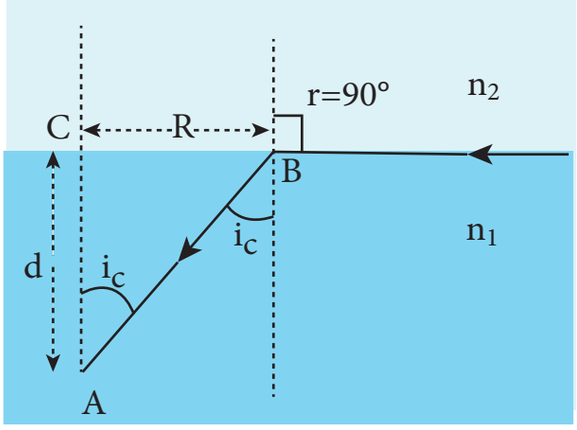


Figure 6.27 Radius of Snell's window

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